

Yue Wu

Division of Applied Mathematics
Brown University, Box F
Providence, RI 02912-9106, USA

Email: yue_wu3@brown.edu
URL: <https://yuewu2002.github.io>

Education

- **Ph.D. Candidate in Applied Mathematics** 09/2023–present
Division of Applied Mathematics, Brown University, Providence, RI 02912, USA
Advisor: *Prof. Chi-Wang Shu*
- **M.Sc. in Applied Mathematics** 09/2023–05/2025
Division of Applied Mathematics, Brown University, Providence, RI 02912, USA
- **B.Sc. in Information & Computational Science** 09/2019–06/2023
School of the Gifted Young, University of Science and Technology of China, Hefei, Anhui 230026, China
- Wuxi No. 1 High School, Wuxi, Jiangsu 214031, China 09/2017–06/2019

Research Interests

- Numerical methods for partial differential equations (PDE)
 - Discontinuous Galerkin (DG) finite element methods
 - Finite difference Weighted Essentially Non-Oscillatory (WENO) methods
 - Structure-preserving numerical methods
- Scientific computing
 - Parallel PDE solver development

Publications and Preprints

1. **Y. Wu**, S. Becker, J.-M. Qiu, and X. Zhang. Nonnegative low-rank matrix correction under an orthogonality constraint in conservative Vlasov simulations. submitted to *J. Sci. Comput.* arXiv:2607.26272
2. **Y. Wu** and C.-W. Shu. A positivity preserving and entropy stable nodal discontinuous Galerkin scheme for ideal MHD. submitted to *Commun. Math. Stat.* arXiv:2604.23885
3. **Y. Wu** and C.-W. Shu. Finite difference alternative WENO schemes with Riemann invariant-based local characteristic decompositions for compressible Euler equations. *J. Comput. Phys.*, 537:114104, Sept 2025. DOI: 10.1016/j.jcp.2025.114104. MR4912873.
4. **Y. Wu** and Y. Xu. A high-order local discontinuous Galerkin method for the p -Laplace equation. *Beijing J. of Pure and Appl. Math.*, 2(1):373–422, Apr 2025. DOI: 10.4310/BPAM.250415002006.

Research Experience

1. **Structure-preserving discontinuous Galerkin schemes for MHD equations**
Brown University 03/2025–04/2026
Supervisor: *Prof. Chi-Wang Shu*
 - Studied all types of explicit structure-preserving conservative schemes for compressible MHD.
 - Developed a nodal discontinuous Galerkin scheme that combines positivity preservation and entropy stability.
2. **Efficient alternative WENO (A-WENO) methods for compressible Euler equations**
Brown University 09/2024–02/2025
Supervisor: *Prof. Chi-Wang Shu*

- Investigated the effect of different transform variables in the local characteristic decomposition on the performance of A-WENO methods.
 - Developed an A-WENO code using Riemann invariants as transform variables to save cost.
3. **Discontinuous Galerkin Methods for the p -Laplace Equation**
 Bachelor's thesis at USTC 12/2022–06/2023
 Supervisor: *Prof. Yan Xu*
- Proved an a priori error estimate for an LDG scheme for the p -Laplace equation.
 - Developed and implemented an efficient preconditioned gradient descent method.
4. **Positivity-Preserving Conservative Low Rank Methods for Vlasov Dynamics**
 Purdue University (remote) 06/2022–08/2022
 Supervisor: *Prof. Xiangxiong Zhang*
- Developed a low-rank correction algorithm with positivity preservation and orthogonality constraints via optimization, which can post-process data from a dynamic low-rank solver.
5. **Numerical Simulation of Plasma Equilibrium Evolution in Nuclear Fusion**
 USTC undergraduate research project 06/2021–05/2022
 Supervisor: *Prof. Mengping Zhang*
- Developed a parallel hybrid finite difference-pseudo spectral code for resistive MHD in toroidal geometry, and performed long-time simulation of resistive tearing mode instability in tokamaks.
 - Checked the results with researchers from the Institute of Plasma Physics, CAS, and against those from existing open-source codes.

Teaching Experience

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| 1. TA: APMA 1655 – <i>Intro. to Prob. & Stats. with Theory</i> , Brown | Spring 2026 |
| 2. TA: APMA 0160 – <i>Intro. to Sci. Computing</i> , Brown | Fall 2025 |
| 3. TA: APMA 1650 – <i>Stat. Inference I</i> , Brown | Spring 2025 |
| 4. TA: APMA 1210 – <i>OR: Deterministic Models</i> , Brown | Fall 2024 |
| 5. TA: <i>Computational Methods B</i> , USTC | Spring 2022 |

Presentations and Workshops

1. Poster session, the 2024 International Congress of Basic Science (ICBS), Beijing, China 07/2024

Professional Services

1. Jorunal reviewer for *J. Comput. Phys.* (since 2025), *J. Sci. Comput.* (since 2025), and *Comput. Math. Appl.* (since 2026)

Honors and Awards

- New Lotus Award, the 2023 SGY Rose Scholarship 06/2024
- USTC Outstanding Undergraduate Award 06/2023
- “Chia-Chiao Lin” Gold Medal in Applied and Computational track & Team Silver Medal & Excellence Prize in Analysis and PDEs track, the 14th S.-T. Yau College Student Mathematics Contest 06/2023
- Gold Prize, USTC Outstanding Student Scholarship 10/2022
- Excellence Prize in Analysis and PDEs track, the 13th S.-T. Yau College Student Mathematics Contest 08/2022
- China National Scholarship 12/2021
- Second Prize, the 13th Chinese Mathematics Competitions 12/2021
- China National Scholarship 12/2020
- Third Prize, USTC Freshman Scholarship 09/2019

Professional Skills

- Programming: MATLAB, C++, Fortran, Python, MPI, OpenMP
- Software: $\text{\LaTeX}$, Mathematica, NGSolve, FEniCS, MFEM, Tecplot
- Language: Mandarin Chinese, English

Extracurricular Activities

- USTC road cycling team member, USTC 09/2019–06/2023
- Monitor of class 2019-3 for math-majored students, SGY, USTC 03/2022–06/2023

last update: August 24, 2026